THE RELATIVE SPEC FUNCTOR F IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Glue-functors criterion. schemes-lemma-glue-functors (cross-chapter, named). Four-step representability template.
- Base change of F. lemma-spec-base-change (carded this batch). Identifies each chart subfunctor with the base-changed affine functor.
- Affine base case. lemma-spec-affine (same chapter, judged non-self-contained this batch, not carded). Represents F over affine opens by Spec of the sections.

Source reference. Original lemma 01LT: sources/stacks-constructions/pinned-original-source.tex, source lines 709–709. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01LS: sources/stacks-constructions/pinned-original-source.tex, source lines 685–685. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01JJ: focused-seeds/geometry/S9-cons-spec/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01JA: focused-seeds/geometry/S9-cons-spec/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00AK: focused-seeds/geometry/S9-cons-spec/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01LM: sources/stacks-constructions/pinned-original-source.tex, source lines 530–530. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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Lemma 0.1. *In Situation 01LM. The functor F is representable by a scheme.*

Proof. We are going to use Schemes, Lemma 01JJ.

First we check that F satisfies the sheaf property for the Zariski topology. Namely, suppose that T is a scheme, that $T = \bigcup_{i \in I} U_i$ is an open covering, and that $(f_i, \varphi_i) \in F(U_i)$ such that $(f_i, \varphi_i)|_{U_i \cap U_j} = (f_j, \varphi_j)|_{U_i \cap U_j}$. This implies that the morphisms $f_i : U_i \to S$ glue to a morphism of schemes $f : T \to S$ such that $f|_{U_i} = f_i$, see Schemes, Section 01JA. Thus $f_i^* \mathcal{A} = f^* \mathcal{A}|_{U_i}$ and by assumption the morphisms φ_i agree on $U_i \cap U_j$. Hence by Sheaves, Section 00AK these glue to a morphism of $\mathcal{O}_T$-algebras $f^* \mathcal{A} \to \mathcal{O}_T$. This proves that F satisfies the sheaf condition with respect to the Zariski topology.

Let $S = \bigcup_{i \in I} U_i$ be an affine open covering. Let $F_i \subset F$ be the subfunctor consisting of those pairs $(f : T \to S, \varphi)$ such that $f(T) \subset U_i$.

We have to show each F_i is representable. This is the case because F_i is identified with the functor associated to U_i equipped with the quasi-coherent $\mathcal{O}_{U_i}$-algebra $\mathcal{A}|_{U_i}$, by Lemma 01LS. Thus the result follows from Lemma 01LT.

Next we show that $F_i \subset F$ is representable by open immersions. Let $(f : T \to S, \varphi) \in F(T)$. Consider $V_i = f^{-1}(U_i)$. It follows from the definition of F_i that given $a : T' \to T$ we have $a^*(f, \varphi) \in F_i(T')$ if and only if $a(T') \subset V_i$. This is what we were required to show.

Finally, we have to show that the collection $(F_i)_{i \in I}$ covers F. Let $(f : T \to S, \varphi) \in F(T)$. Consider $V_i = f^{-1}(U_i)$. Since $S = \bigcup_{i \in I} U_i$ is an open covering of S we see that $T = \bigcup_{i \in I} V_i$ is an open covering of T. Moreover $(f, \varphi)|_{V_i} \in F_i(V_i)$. This finishes the proof of the lemma. $\square$